

6 Discussion

6.1 Conclusions

In this work, we revisited the Blue-Eyed Islander riddle with the purpose of generalizing the findings with respect to the visibility configurations. In order to do this, we introduced more general Kripke models that can capture partial visibility. Using this theory, we implemented a method based on these models to check convergence under different visibility constraints. Our findings provide insights on different topics. Firstly, the results indicate that the description of the generalized Kripke model is accurate and complete. Secondly, our findings indicate that under partial visibility configurations, convergence may be delayed and even totally prevented. In other words, it is possible that it will require more than k days for k blue-eyed islanders to leave. Moreover, some configurations make it impossible for some blue-eyed agents to ever leave. Through the non-converging cases, it was possible to spot common patterns that prevent convergence. These include blind blue-eyed agents, blind spots between two agents, and sparse visibility in general. Finally, the key takeaway is that uniform solvability requires perfect vision for all agents. All in all, our results demonstrate how even small deviations from common assumptions such as perfect visibility can alter epistemic dynamics.

6.2 Relation with Other Puzzles and Implementations

Our framework and introduction of visibility matrices connect greatly to the underlying structures of many other riddles in the same field. This is because, although many of these riddles differ in context and content, our visibility matrix on which we base our Kripke models provides a uniform structure to represent these different scenarios. All riddles share that agents reason about what others can or cannot distinguish based on their visibility and observations.

An example of such a riddle is the Hats-On-A-Line puzzle (Aravamuthan & Lodha, 2006). In this puzzle, each agent has a black or white hat, however they do not know what colour their hat is. Agents are forced to stand in a line where each agent can only see those in front of them. These visibility configurations can be represented by the visibility matrices

$$E = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & \dots \\ 0 & 0 & 1 & 1 & 1 & \dots \\ 0 & 0 & 0 & 1 & 1 & \dots \\ 0 & 0 & 0 & 0 & 1 & \dots \\ 0 & 0 & 0 & 0 & 0 & \dots \\ \dots & \dots & \dots & \dots & \dots & 0 \end{bmatrix} . \quad \text{This is one of many examples where our generalized approach}$$

can capture observational patterns in a riddle (Van Ditmarsch & Kooi, 2015). Consequently, it may be used in combination with the necessary semantics and additional structures to create solvers for these riddles. That is why we believe that our method is a strong backbone to analyse and compare a wide range of logical puzzles.

6.3 Future Research

There are several directions that possible future implementations could take. First of all, a possible improvement could be to dive deeper into the characterization or mapping of the different visibility matrices. We mentioned a few of these observations, however perhaps there are more underlying (and deeper) properties that may become apparent when analysing more thoroughly. This could lead to general theorems about epistemic solvability under partial visibility. Secondly, the project can be extended on a larger scale, where the behaviour is analysed under larger sets of agents, which would be beyond the scope of this work regarding complexity and computational

resources. Moreover, the puzzle could be extended by adding more eye colours to the world. This could be possible by adding the eye colours as additional propositional atoms. Finally, our implementation generalizes the riddle with respect to the visibility. Future works could focus on generalizing other core assumptions of the riddle, for example by adding unreliable announcements or by limiting reasoning capabilities for some agents.